

The following first-level items should be included in the IE 492 Final Report.

- Abstract (in English and Turkish)
- 1. Introduction
 - a. A summary of the IE problem handled
 - b. A short overview of identification, analysis and solution methodologies
 - c. Improvements (realized or expected) due to your work
 - d. Brief summary of conclusions
 - e. Contents of the report
- 2. Problem Definition, Requirements and Limitations
 - a. What seems to be the problem?
 - b. What is done to understand the causes of the problem?
 - c. Needs and requirements of the system/customer
 - d. Limitations and constraints (including important environmental/social/legal/ethical/geopolitical considerations)
 - e. Data gathered and used in the identification phase
 - f. A context diagram (a systemic view) of the handled design problem
 - g. Performance criteria and potential improvements
- 3. Analysis for Solution/Design Methodology
 - a. Literature overview
 - b. Alternative solution/design approaches
 - c. Assumptions
 - d. Brief overview of the selected approach(es)
 - e. IE skills/tools/techniques/methods to be integrated to implement the proposed methodology
- 4. Development of Alternative Solutions
 - a. Detailed technical description of the design process for generating meaningful alternatives
- 5. Comparison of Alternatives and Recommendation
 - a. Numerical studies or evaluation procedure
 - b. Proposing a promising solution along with justification in terms of predefined criteria
 - Limitations
 - Sensitivity
 - c. Further assessment of the recommended solution
 - Is the solution consistent with system requirements and limitations?
 - Can the solution be implemented?
 - Is the solution sustainable? What are the data/management requirements?
 - Is the solution robust?
- 6. Suggestions for a Successful Implementation
 - a. How can the design be implemented?
 - b. Can the design be integrated with the overall system?
 - c. How often should any solution obtained be revised?
- 7. Conclusions and Discussion
 - a. Summarize IE tools, techniques, methods that are integrated towards a successful completion of the project
 - b. Summarize the merits and significance of your design
 - c. Economic, environmental, ethical and societal impacts of your design
- References
- Appendix (if necessary)